1 Cross-Modal Uncertainty and Relative Geometric Reliability

I am interested in a distinction between **using uncertainty as a measurement weight** and **understanding whether uncertainty is meaningfully comparable across sensing modalities**.

Let

$$U_R,\ U_T$$

denote the predicted geometric uncertainties associated with RGB and thermal measurements, respectively, and let

$$E_R,\ E_T$$

denote their corresponding actual geometric errors.

1.1 Relationship to Existing Work

Existing uncertainty-aware visual odometry methods largely establish the usefulness of modality-specific uncertainty.

For RGB-based visual odometry, such as MAC-VO, the relationship can be represented conceptually as

$$U_R \longrightarrow E_R,$$

where the estimated uncertainty indicates how much an RGB geometric measurement should be trusted. The resulting covariance can then be propagated and used during visual odometry optimization.

Thermal extensions investigate the analogous relationship

$$U_T \longrightarrow E_T,$$

where uncertainty associated with thermal measurements is used to estimate their geometric reliability.

Multimodal approaches can subsequently use both uncertainty estimates for adaptive measurement weighting:

$$(U_R, U_T) \longrightarrow \text{measurement weighting/selection} \longrightarrow \text{VO}.$$

Therefore, simply using RGB and thermal uncertainty to dynamically weight measurements is not, by itself, the research contribution I am interested in.

1.2 Research Question

I instead want to investigate whether modality-specific uncertainty provides a meaningful **cross-modal measure of relative geometric reliability**.

Conceptually, I want to determine whether

$$\boxed{\frac{U_R}{U_T} \overset{?}{\longleftrightarrow} \frac{E_R}{E_T}}$$

where the ratios are illustrative rather than necessarily the final mathematical representation.

More generally, the question is whether the relationship

$$f(U_R, U_T)$$

is predictive of

$$g(E_R, E_T),$$

where $f(\cdot)$ represents a measure of relative predicted uncertainty and $g(\cdot)$ represents relative actual geometric error.

For example, if

$$U_R > U_T,$$

does this reliably imply

$$E_R > E_T?$$

In other words, when the uncertainty estimates indicate that thermal is more reliable than RGB, is thermal actually providing more accurate geometric information?

1.3 Why Cross-Modal Comparability Is Not Guaranteed

Even if U_R and U_T are individually useful uncertainty estimates, their numerical values are not necessarily directly comparable.

The two estimates may arise from different sensing physics, feature characteristics, error distributions, network behavior, calibration properties, and environmental sensitivities. Therefore,

$$U_R < U_T$$

does not automatically guarantee

$$E_R < E_T.$$

This motivates investigating whether RGB and thermal uncertainties form a common, calibrated measure of geometric reliability across environmental conditions.

The central question can therefore be stated as:

> **Are modality-specific geometric uncertainty estimates cross-modally comparable and predictive of relative geometric reliability between RGB and thermal?**

1.4 Experimental Interpretation

I would first evaluate the modality-specific relationships [$U_R \leftrightarrow E_R$ and $U_T \leftrightarrow E_T$] to determine whether each uncertainty estimate is individually calibrated with geometric error.

I would then investigate the more important cross-modal relationship $[(U_R, U_T) \longleftrightarrow (E_R, E_T),]$ with particular emphasis on whether relative uncertainty predicts relative geometric accuracy.

This relationship should also be evaluated across environmental regimes such as low illumination, low texture, poor thermal contrast, and other visually degraded conditions.

1.5 Potential Outcomes

There are several meaningful possible outcomes.

Cross-modally comparable uncertainty. If relative uncertainty strongly predicts relative geometric error, then $[U_R < U_T \quad \Rightarrow \quad E_R < E_T]$ with sufficiently high consistency. In this case, the uncertainty relationship could directly support adaptive modality weighting or selection.

Informative but miscalibrated uncertainty. The uncertainty estimates may contain useful relative reliability information without being directly comparable. A cross-modal calibration or reliability model could then be introduced: $[h(U_R, U_T, \mathcal{C}) \longrightarrow P(E_R < E_T),]$ where $(\mathcal{C})$ represents environmental or temporal context.

Poor cross-modal comparability. The independently estimated uncertainties may fail to predict which modality is geometrically more reliable. This would motivate investigating the conditions under which comparability fails and whether a different representation of cross-modal reliability is necessary.

1.6 Potential VO Extension

If a meaningful relationship is established, I can then investigate whether explicitly modeling it improves adaptive RGB–thermal visual odometry:

[RGB/Thermal measurements $\longrightarrow (U_R, U_T) \longrightarrow$ cross-modal reliability model $\longrightarrow$ adaptive weighting/selection $\longrightarrow$ VO.]

The resulting research question is therefore not simply whether uncertainty can be used for multimodal fusion. Instead, I am asking whether independently estimated RGB and thermal uncertainties provide a meaningful **common reliability language across modalities**, and whether explicitly modeling that relationship can reduce visual-odometry pose error and improve robustness under changing sensing conditions.